Hyung Cho

AI Evaluation | LLM and Retrieval Evals, Experiments, and Root-Cause Investigation

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SUMMARY

Builds quantitative evaluations for LLM and retrieval systems: test sets, scheduled regression checks, LLM-judge validation, and side-by-side comparisons of models and rubrics. Spent 10 years as a validation engineer at Zebra Technologies isolating root causes across hardware, firmware, software, and test infrastructure, and writing the final reports for 6 shipped products. Works in Python and SQL, with evals automated in GitHub Actions.

EXPERIENCE

Independent AI Engineer | Self-directed projects, Remote **Jun 2025 - Present**

Solo product work, built by directing AI coding agents (Claude Code, Codex).

Purvia, permission-aware RAG and support-bot app, open source

- Traced a 27% false-resolve rate (5% ceiling) in an escalation eval to the offline LLM judge, not the product, which made 0 false resolves in 55 end-to-end runs; compared five judge designs on replayed drafts and kept the one that cut wrong verdicts from 38 of 280 to 0 of 300.
- For the live answer gate, compared a stricter value-extraction rubric against a stronger judge model on 85 stored drafts; the rubric let 1 of 66 should-escalate drafts through against 15 of 66, for about $0.00007 more per reply.
- Built a 60-question retrieval benchmark that scores vector, keyword, and hybrid search on recall@k, MRR, and nDCG@5 and posts the difference from main on pull requests; it flagged a 0.510 recall@5 drop on a deliberately broken change.
- Traced a question the RAGAS coverage gate flagged as retrieving nothing to a keyword-stemming mismatch, and added a stem-prefix search tier that raised hybrid recall@5 from 0.950 to 0.967 with no metric worse.
- Designed an escalation eval: an 18-question test set whose deterministic leak checks block pull requests, a scheduled LLM-judged check against the 5% ceiling, and a check that the offline and runtime judges agree.
- Added a scheduled RAGAS job (faithfulness, answer relevancy, context precision, context recall) across two permission levels, with coverage and API-error gates and a cross-check against a separate Claude judge.
- After a permissions benchmark published 0.000 for 42 runs because its test users lacked a workspace membership, added guard tests so the evals refuse to report a score when they measured nothing.
- Built a 10,000-chunk benchmark that sweeps HNSW search settings for users who can see 50%, 10%, and 1% of the data, with scheduled jobs that commit dated results and open an issue when a run fails or measures nothing.
- Wrote access control into the Postgres vector and keyword retrieval functions, backed by row-level security across 48 SQL migrations, so every search returns only chunks the signed-in user may see.

Rep Today, native iOS workout app, pre-release

- Instrumented 13 pre-registered funnel events through a Convex HTTP endpoint with opt-out consent and rate limits, and verified accepted, rejected, rate-limited, and opted-out behavior in production checks.

Volunteer AI/ML Engineer | Omdena, Remote **Apr 2023 - Jun 2025**

Unpaid collaborative ML projects with global volunteer teams.

- Fine-tuned open-source LLMs (LLaMA, Falcon) and iterated on chatbot prompts with an evaluation workflow.

Career break and ML retraining **Aug 2018 - Mar 2023**

Self-directed ML study and the FourthBrain ML Engineering bootcamp (2022); 29 public ML projects on GitHub.

Senior Validation Engineer | Zebra Technologies, Agoura Hills, CA **Feb 2008 - Jul 2018**

- Isolated root causes of failures across hardware, firmware, software, and test infrastructure, and proved the fixes.
- Led product validation for 6 shipped hardware products: test plans, procedures, acceptance criteria, and final reports.

SKILLS

AI evaluation: LLM evaluation, Model evaluation, LLM-as-judge, RAGAS, Retrieval metrics, Prompt engineering
Data and analysis: Python, SQL, PostgreSQL, Pandas, Data analysis, Statistics, Product analytics
Retrieval and ML: RAG, Hybrid retrieval, Reranking, pgvector, Machine learning, Deep learning, Computer vision
Investigation and tooling: Failure analysis, Regression testing, GitHub Actions, LangSmith, OpenAI API, Anthropic API

EDUCATION AND TRAINING

B.S., Electrical Engineering, California Polytechnic State University, San Luis Obispo
FourthBrain ML Engineering bootcamp (2022); DeepLearning.AI deep learning and generative AI courses; Google Advanced Data Analytics Professional Certificate